Light

GCSE Computer Science – Algorithms and Trace Tables (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which of the following best describes an algorithm?

- ○ **A.** A sequence of steps that can be followed to complete a task
- ○ **B.** A piece of hardware inside the computer
- ○ **C.** A type of programming language
- ○ **D.** A file saved on the hard drive

Q2 (1 mark)

What is the relationship between an algorithm and a computer program?

- ○ **A.** They are exactly the same thing

- ○ **B.** A computer program is an implementation of an algorithm in a specific programming language
- ○ **C.** An algorithm is always faster than a program
- ○ **D.** Programs do not use algorithms

Q3 (1 mark)

What does the term 'decomposition' mean in computational thinking?

- ○ **A.** Deleting old files from a computer
- ○ **B.** Making a program run faster
- ○ **C.** Breaking a problem into smaller, more manageable sub-problems
- ○ **D.** Converting code from one language to another

Q4 (1 mark)

What does 'abstraction' mean in the context of computational thinking?

- ○ **A.** Removing unnecessary detail from a problem to focus on what is important
- ○ **B.** Adding more detail to make a solution clearer
- ○ **C.** Writing code in abstract art style
- ○ **D.** Making a problem more complex

Q5 (1 mark)

What is the main purpose of using a trace table?

- ○ **A.** To make code run faster
- ○ **B.** To draw flowchart diagrams
- ○ **C.** To encrypt data for security
- ○ **D.** To track how variable values change as an algorithm executes step by step

Q6 (1 mark)

In a flowchart, which shape represents a decision or condition?

- ○ **A.** Rectangle
- ○ **B.** Oval
- ○ **C.** Parallelogram
- ○ **D.** Diamond

Q7 (1 mark)

In this algorithm, which line represents the INPUT?

```
Line 1: num ← USERINPUT
Line 2: result ← num * 2
Line 3: OUTPUT result
```

- ○ **A.** Line 1

- ○ **B.** Line 3
- ○ **C.** Line 2
- ○ **D.** All lines are inputs

Q8 (1 mark)

What is the value of y after this code executes?

```
x ← 5
y ← x * 3
x ← 10
```

- ○ **A.** 5
- ○ **B.** 10
- ○ **C.** 15
- ○ **D.** 30

Q9 (1 mark)

When does a WHILE loop stop executing?

- ○ **A.** After exactly 10 iterations
- ○ **B.** When the user presses a key
- ○ **C.** When the condition becomes FALSE
- ○ **D.** When it runs out of memory

Q10 (1 mark)

How many times does the loop body execute?

```
FOR i ← 1 TO 4
    OUTPUT i
ENDFOR
```

- ○ **A.** 3 times
- ○ **B.** 5 times
- ○ **C.** 1 time
- ○ **D.** 4 times

Q11 (1 mark)

In an IF-THEN-ELSE statement, when is the ELSE branch executed?

- ○ **A.** Always, after the THEN branch
- ○ **B.** Only when the condition is TRUE
- ○ **C.** At random
- ○ **D.** Only when the condition is FALSE

Q12 (1 mark)

In AQA pseudocode, what does the symbol ← mean?

- ○ **A.** Assignment - stores a value in a variable

- B. Less than or equal to
- C. Points to a memory address
- D. Moves data to the left

Q13 (2 marks)

Trace this algorithm:

```
total ← 0
FOR n ← 1 TO 4
    total ← total + n
ENDFOR
```

What is the final value of total?

- A. 4
- B. 6
- C. 0
- D. 10

Q14 (2 marks)

Trace this algorithm:

```
x ← 20
count ← 0
WHILE x > 5
    x ← x - 4
    count ← count + 1
ENDWHILE
```

What is the final value of count?

- A. 5
- B. 3
- C. 4
- D. 6

Q15 (2 marks)

Trace this algorithm with input value 25:

```
INPUT score
IF score >= 70 THEN
    grade ← "A"
ELSE
    IF score >= 50 THEN
        grade ← "B"
    ELSE
        grade ← "C"
    ENDIF
ENDIF
```

What is the value of grade?

- A. A
- B. B
- C. C
- D. 25

Q16 (2 marks)

What does this algorithm determine?

```
INPUT number
IF number MOD 2 = 0 THEN
    OUTPUT "Type A"
ELSE
    OUTPUT "Type B"
ENDIF
```

- ○ **A.** Whether number is positive or negative
- ○ **B.** Whether number is even or odd
- ○ **C.** Whether number is greater than 2
- ○ **D.** Whether number is a prime number

Q17 (2 marks)

Trace this algorithm with input 47:

```
INPUT num
tens ← num DIV 10
units ← num MOD 10
```

What are the values of tens and units?

- ○ **A.** tens = 4, units = 7
- ○ **B.** tens = 7, units = 4
- ○ **C.** tens = 47, units = 0
- ○ **D.** tens = 10, units = 37

Q18 (2 marks)

Given the array `names ← ["Ali", "Ben", "Cat", "Dan"]`, what does `OUTPUT names[2]` display?

- ○ **A.** Ali
- ○ **B.** Ben
- ○ **C.** Cat
- ○ **D.** Dan

Q19 (2 marks)

What does this algorithm find?

```
biggest ← data[0]
FOR i ← 1 TO LEN(data) - 1
    IF data[i] > biggest THEN
        biggest ← data[i]
    ENDIF
ENDFOR
```

- ○ **A.** The first element of the array
- ○ **B.** The last element of the array
- ○ **C.** The maximum (largest) value in the array
- ○ **D.** The sum of all elements

Q20 (2 marks)

Trace this algorithm with input "HELLO":

```
INPUT word
length ← LEN(word)
last ← word[length - 1]
```

What is the value of last?

- A. H
- B. E
- C. L
- D. O

Q21 (2 marks)

Evaluate this Boolean expression when a = 5 and b = 12:

```
(a < 10) AND (b > 10)
```

- A. TRUE
- B. FALSE
- C. 5
- D. 12

Q22 (2 marks)

Trace this algorithm:

```
first ← "Hello"
second ← "World"
message ← first + " " + second
```

What is the value of message?

- A. HelloWorld
- B. Hello World
- C. first second
- D. Hello " " World

Q23 (2 marks)

What does this algorithm count in the array scores?

```
count ← 0
FOR i ← 0 TO LEN(scores) - 1
    IF scores[i] < 40 THEN
        count ← count + 1
    ENDIF
ENDFOR
```

- A. How many scores equal 40
- B. The total of all scores
- C. How many scores are below 40
- D. The average score

Q24 (3 marks)

Trace this nested loop algorithm:

```
result ← 0
FOR i ← 1 TO 2
    FOR j ← 1 TO 3
        result ← result + 1
```

```
    ENDFOR
ENDFOR
```

What is the final value of result?

- A. 3
- B. 5
- C. 6
- D. 2

Q25 (3 marks)

Trace this algorithm with array [10, 15, 20, 25, 30]:

```
FOR i ← 0 TO 4
    IF nums[i] > 18 THEN
        nums[i] ← nums[i] - 5
    ENDIF
ENDFOR
```

What is the final state of the array?

- A. [5, 10, 15, 20, 25]
- B. [10, 15, 15, 20, 25]
- C. [10, 15, 20, 25, 30]
- D. [10, 10, 15, 20, 25]

Q26 (3 marks)

Analyse what this algorithm produces for input "CAT":

```
INPUT word
result ← ""
FOR i ← LEN(word) - 1 DOWNTO 0
    result ← result + word[i]
ENDFOR
OUTPUT result
```

- ○ **A.** CAT
- ○ **B.** TAC
- ○ **C.** CATCAT
- ○ **D.** AAA

Q27 (3 marks)

Trace this algorithm with inputs x = 8, y = 3:

```
INPUT x, y
IF x > y THEN
    IF x MOD y = 0 THEN
        result ← "divisible"
    ELSE
        result ← "not divisible"
    ENDIF
ELSE
    result ← "x not larger"
ENDIF
```

What is the value of result?

- ○ **A.** divisible
- ○ **B.** x not larger
- ○ **C.** 8

- ○ **D.** not divisible

Q28 (3 marks)

Trace this algorithm:

```
product ← 1
FOR n ← 2 TO 5
    product ← product * n
ENDFOR
```

What is the final value of product?

- ○ **A.** 14
- ○ **B.** 120
- ○ **C.** 20
- ○ **D.** 5

Q29 (3 marks)

Analyse what this algorithm does:

```
position ← -1
FOR i ← 0 TO LEN(list) - 1
    IF list[i] = target THEN
        position ← i
    ENDIF
ENDFOR
```

- ○ **A.** Finds the first position of target in the list
- ○ **B.** Finds the last position of target in the list

- ○ **C.** Counts how many times target appears
- ○ **D.** Removes target from the list

Q30 (3 marks)

Consider this 2D array and algorithm:

```
grid ← [[1, 2, 3],
        [4, 5, 6],
        [7, 8, 9]]
rowTotal ← 0
FOR col ← 0 TO 2
    rowTotal ← rowTotal + grid[1][col]
ENDFOR
```

What is the final value of rowTotal?

- ○ **A.** 6
- ○ **B.** 12
- ○ **C.** 15
- ○ **D.** 45

Total: **55** marks • Time: **30 min**

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